

Paper bibliography:

Josué Alonso Yee-Duarte, Ilie S. Racotta, Marian Alejandra Camacho-Mondragón, Nefertiti Taydé Roldán-Wong, Diana Patricia Carreño-León, Evgueni Shumilin, Karen A. Kidd, Marcial Arellano-Martínez. Contrasting reproductive health of female clams *Megapitaria squalida* from two nearby metal-polluted sites in the Gulf of California: Potential effects of copper, lead, and cobalt. *Marine Pollution Bulletin*. Volume 160. 2020. 111583. ISSN 0025-326X. <https://doi.org/10.1016/j.marpolbul.2020.111583>.

**Quote 1:** “As regards the assessment of energy content, we examined 54 clams from Santa Rosalia. ... Those clams measured between 5.9 and 7.3 cm ( $6.7 \pm 0.05$  cm) in shell length and between 51.8 and 109.6 g ( $78.6 \pm 1.8$  g) in total weight. For San Lucas, ... Those clams measured between 5.7 and 7.8 cm ( $6.8 \pm 0.06$  cm) in shell length and between 46.6 and 152.6 g ( $87.5 \pm 2.8$  g) in total weight. There was no significant difference in sizes between sites ( $t_{109} = -1.21$ ,  $P = 0.22$ ); however, the clams of Santa Rosalia were lighter ( $t_{109} = -2.51$ ,  $P = 0.01$ ) than those from San Lucas.”

**Issues:** Despite the language and heavy reliance on p values in describing the clam size differences, I think the presentation of these statistics is actually quite strong. They gave the measured shell lengths, along with mean and SEM, and those were near identical in the two sites, which they were able to describe as no significant difference, citing a T score and P value. Same goes for the weight of the clams, and gave the sign of change in words to accompany the P value (e.g. they didn't just say the weight of clams at the two sites were significantly different, they said Santa Rosalia were lighter than San Lucas, indicating directionality)

**Improvement:** The only improvement would be to use clarity language to describe the size difference. E.g. “There was no clear statistical difference in sizes between sites”.

**Quote 2:** “The digestive gland and gills contained the highest concentrations of most metals (except for Cu, mantle > gills in both sites), followed by the mantle, ovary, and muscle. However, only Cu and Co were consistently higher in all tissues of clams from Santa Rosalia versus San Lucas, especially Cu in the digestive gland ( $226.2 \text{ mg kg}^{-1}$ ) and Co in the ovary ( $2.1 \text{ mg kg}^{-1}$ ). Clams from Santa Rosalia also showed higher Mn and Zn levels in most tissues, except for the digestive gland, and higher Pb levels in the digestive gland ( $2.7 \text{ mg kg}^{-1}$ ) and the ovary ( $1.04 \text{ mg kg}^{-1}$ ). In contrast, U was slightly higher in most tissues of clams from San Lucas, except muscle with clams from Santa Rosalia having higher concentrations ( $0.24 \text{ mg kg}^{-1}$ )”

**Issues:** They don't actually use any statistics to describe the differences in metal concentrations in any of the sampled tissues, instead using vague words to describe the observed trends. However, upon investigation, I believe that this is because  $n=1$  for each site, so they could not carry out substantial statistical tests which would support the claims being made.

**Improvement:** Like was said in class, if there is poor experimental design there is not much you can do statistically. While these results are certainly interesting, the low sample number limits the amount of power you can have.

**Quote 3 (this accompanies a table that has the p values and t-test scores in it):** “In the mantle, only carbohydrate concentrations were significantly lower in clams from Santa Rosalia”

**Issues:** While it is true that the mantle only had one significantly different biochemical when looking at the p value (carbohydrates  $p = 0.001$ , lipids  $p = 0.140$ , protein  $p = 0.079$ ), the wording of the sentence makes it seem as if there are no differences between the lipid and protein concentrations in the mantle of clams at the two sites.

**Improvement:** “In the mantle, carbohydrate concentrations were significantly lower in clams from Santa Rosalia, while it is statistically unclear if lipids and protein concentrations in the mantle are also strongly correlated with site.”